

tRansparent *deep/machine*^[B] learners for Official Statistics

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Motivation: main principle

- (Bayesian/) **deep learning**

(B-NN, GAN, AE, VAE, ... LLM: transformer-architectures with encoder and/or decoder blocks)

to add: → *measures of uncertainty*

In the same way as:

- *point.est. of classical mathematical statistics*

→ *confidence/ credible intervals*

where uncertainty measures are built by using:

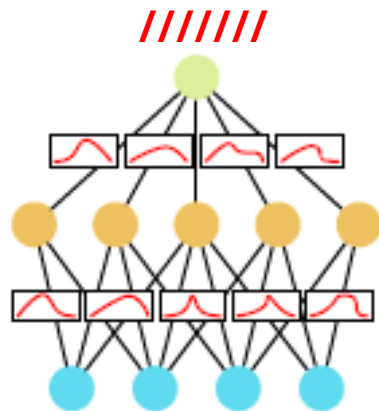
*exact or approximative **methods***

Examples of methods for measuring **uncertainty** of Deep/M.L.

- **Analytic tools** to understand/evaluate deep learning models:

[(2018) *Deep Neural Networks as Gaussian Processes*, Lee, J. etal, <https://arxiv.org/abs/1711.00165>
https://en.wikipedia.org/wiki/Neural_network_Gaussian_process#cite_note-:1-4

- Large-scale evaluation of multiple **uncertainty estimation techniques** applied to various DL/LLMs and tasks



from: <https://arxiv.org/abs/2007.06823>

[(2024) *Look Before You Leap: An Exploratory Study of Uncertainty Measurement for Large Language Models*,
Huang, Y. etal, <https://arxiv.org/abs/2307.10236>]

E.g.: Question → DL/LLM →

- M1: **single** inference (max/avg **prob.** or max/avg **entropy**) → Answer A1
- M2: **Stochastic** inference/sample based/**model** variation (e.g. dropout, deep ensembles) (**VR**, **VR0**) → Answers A2.1, A2.2, ...
- M3: **Stochastic/Data** Perturbation (maxDiff **VR**, MaxDiff **VR0**) → Answers A3.1, A3.2, ...

- Using **Bayesian methods** to:

- mitigate risks arising from overly confident yet incorrect predictions made by LLMs
- provide uncertainties over predictions, which can enrich decision-making
- enable the use of domain-knowledge priors

[(2024) *Position: Bayesian Deep Learning is Needed in the Age of Large-Scale AI*, Papamarkou, Th. Etal, <https://arxiv.org/abs/2402.00809>]

automatic, Deep Learner: a good example

autoBNN :

„**interpretable** time series forecasting models

high-quality **uncertainty estimates**

scales effectively for use on large datasets.

It combines the interpretability of traditional **probabilistic approaches** with the scalability and flexibility of neural networks.“

*It uses **GP** ⇔ **BNN***

and Dirichlet priors for the weights which are learnable.

<https://research.google/blog/autobnn-probabilistic-time-series-forecasting-with-compositional-bayesian-neural-networks/>

Examples at Statistics Iceland : GP and (B)NN

- *ts (conditional-) forecasting* with GAMs and **GP** priors:

Bayes+unsupervised+ prior knowledge & the equiv. **NN**

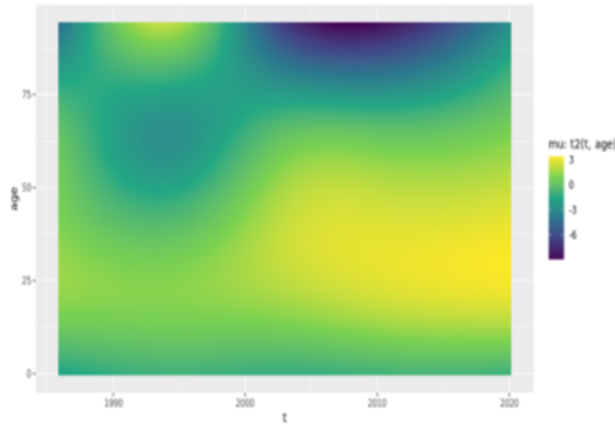
- *SDC – Bayesian risk* measures
- *synthetic* data (GANs, NNs, &)
- unsupervised *anomaly* detection for data privacy (Bayesian NN, GANs , VAE; D.i.f.)
- *differential privacy & Deep Learners*

E1. ts

Simple view on Bayesian modeling and forecasting

- *Posterior_distribution* ~
- *Prior_distribution* * *Likelihood_data*
- |
- | *greater role in model fitting*
- *greater role in forecasting*

HGAM & R-tools



$$response \sim P(f(\dots(f(\eta))))$$

$$\eta = \beta \mathbf{x} + \gamma \mathbf{z} + \sum s_k(a_k, by = c) + \dots$$

|

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smooth dependencies (including *GP*)

group level (cross-/ clustering) attributes and effects

population level attributes and effects

or **unknown form/shape**

GP : (as prior probability distribution over functions) :: connection between Machine Learning and Statistical Modeling :: it learns from data the *strength* of the relation between observations (not the shape)

Implementation: **R-tools** ([*brms*](#), [*mvgam*](#) and [*mgcv*](#) packages)

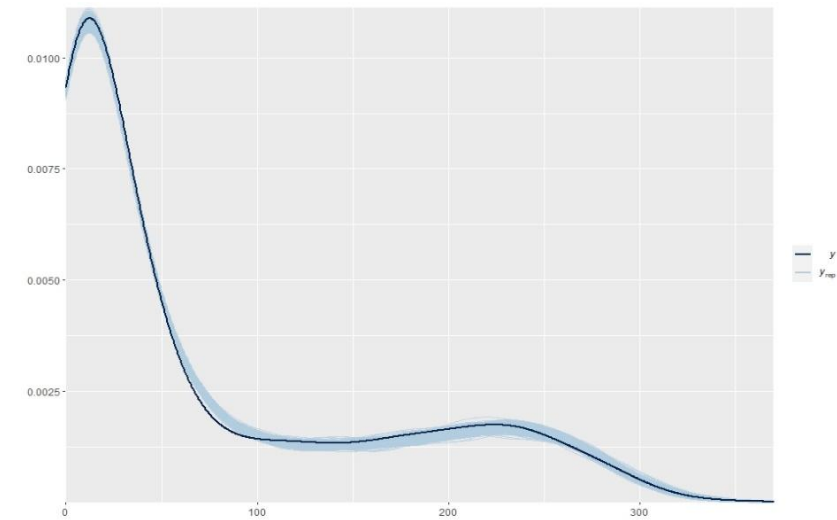
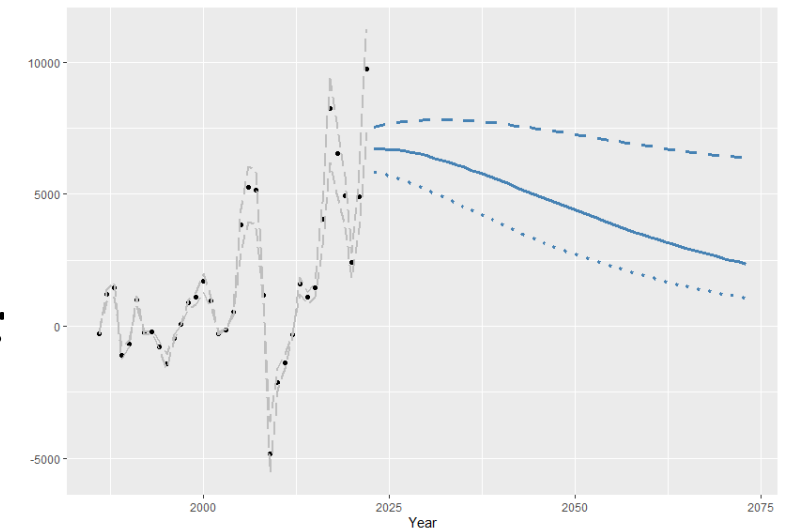
can include: [*me*](#)(...), [*mi*](#)(...), [*mi & me*](#), [*mo*](#)(...), [*cs*](#)(...), [*autocor*](#)(...)

posterior distribution checks, LOO-validation, ...

Special component

Advantages of modelling with Gaussian Processes (GP):

- rich structure of the covariance function capturing *quasi-periodicity, non-stationarity, complex interactions & trend, periodicity, seasonality, autoregressive character* that affect the model predictions in both inter/extrapolation
- good *forecasting* performance
- *related/equivalent* to: State Space models, kriging, **Bayes-Neural Networks**



E2. SDC

Bayes and the privacy budget: risk measures, prior and posterior prob.

DP & classical DC & network ideas

Under adversary model & extra-knowledge, for an arbitrary unit:

$$\text{prior}_{id} = P(\text{discl}_{id}) = p_0$$

$$\text{prior}_{attr} = P(\text{discl}_{attr}) = q_0$$

„Prior-itizing Privacy: A Bayesian Approach to Setting the Privacy Budget in Differential Privacy”
(Kazan and Reiter, 2024)

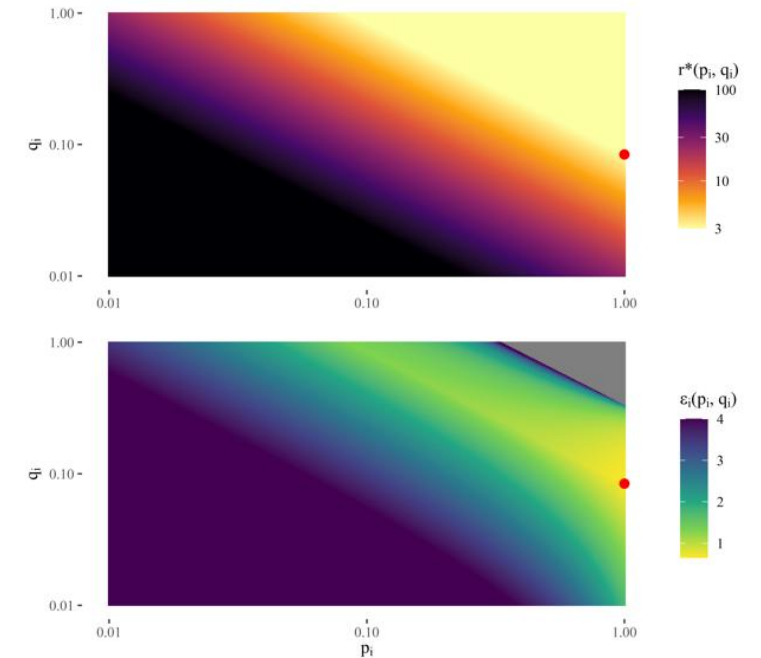
Relative disclosure risk as posterior/prior ratio --->

optimum curve $\text{risk} = F(\text{discl}_{id}, \text{discl}_{attr})(\text{unit})$,

for all units from bounds of (w.r.t. the *risk-profile*):

$$\frac{P(\text{discl}_{id}, \text{discl}_{attr} | \text{output released})}{P(\text{discl}_{id}, \text{discl}_{attr})} =$$

$$\frac{P(\text{discl}_{attr} | \text{discl}_{id}, \text{output released})}{P(\text{discl}_{attr} | \text{discl}_{id})} \cdot \frac{P(\text{discl}_{id} | \text{output released})}{P(\text{discl}_{id})}$$



R-tools:

<https://github.com/sdcTools>

but also:



- *R* -Synthetics:

NP BayesImputeCat $_{NP}^B$, synMicrodata $_{NP}^B$, synthpop, **RGAN**, *synthesizer*

- *R*-Differential Privacy:

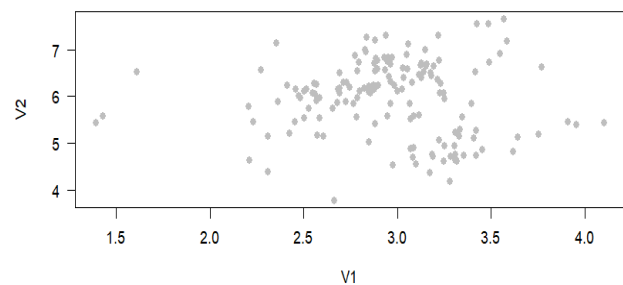
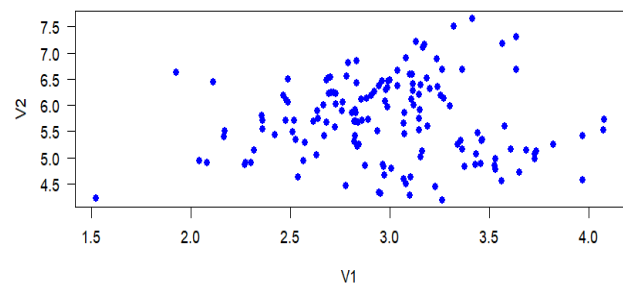
Dppack, **diffpriv** $_{RDP}$

- *R* – Anomalies:

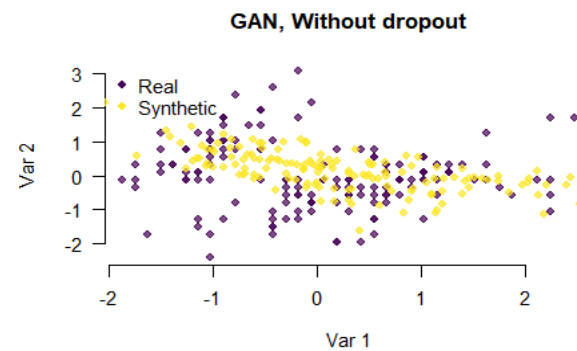
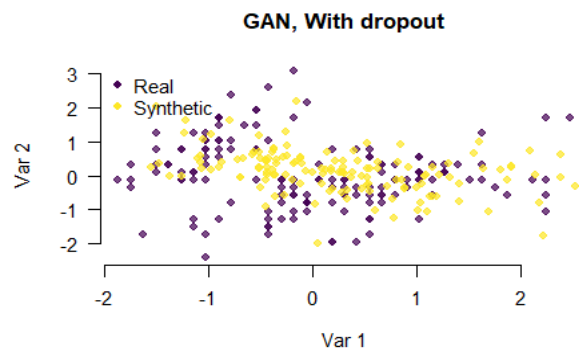
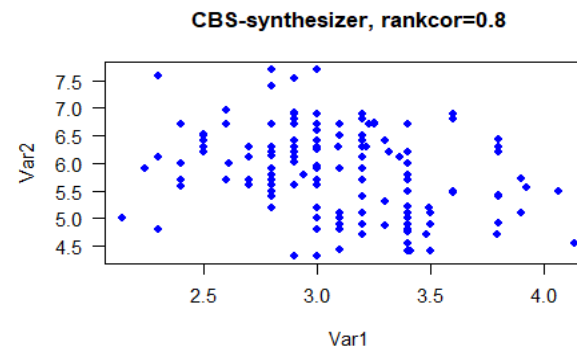
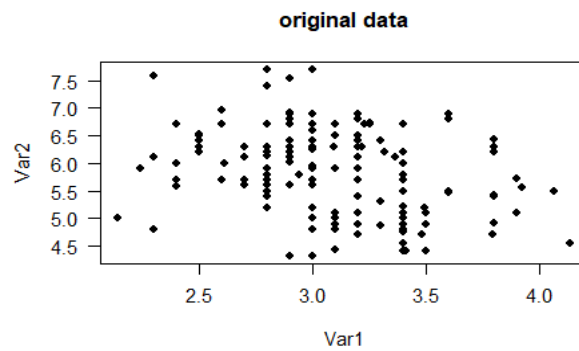
HRTnomaly (*Bayes*HRT, **dif**), keras(**AE**), h2o(**AE**)

Synthetics

Bayesian synthetics

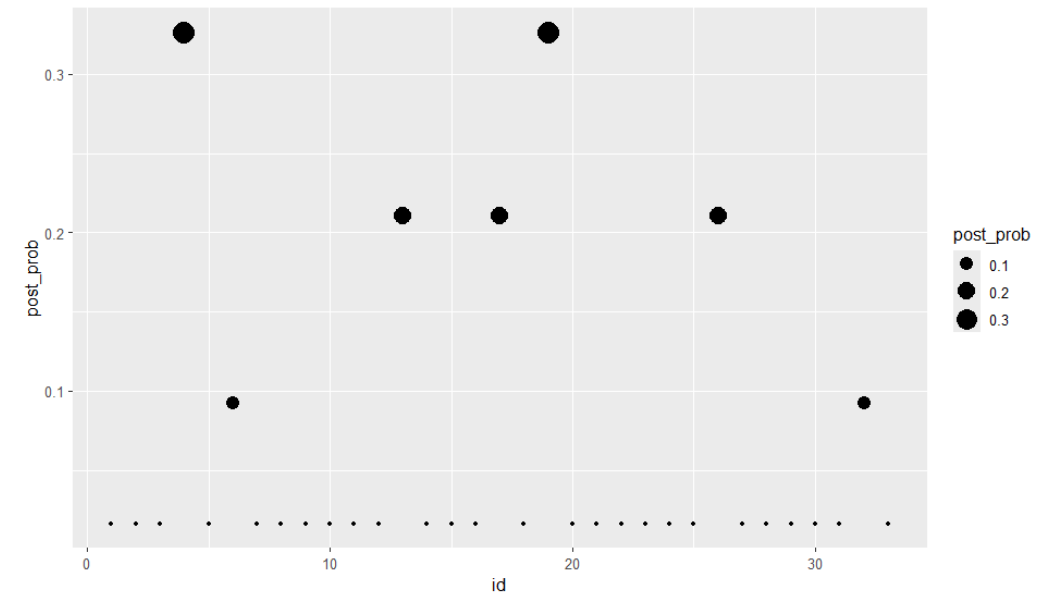
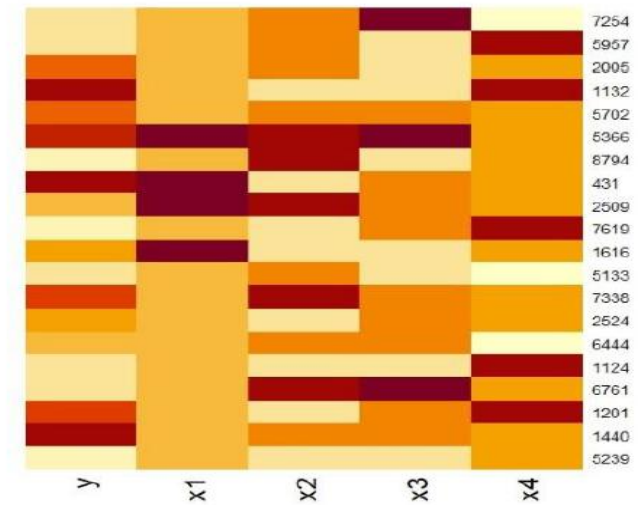
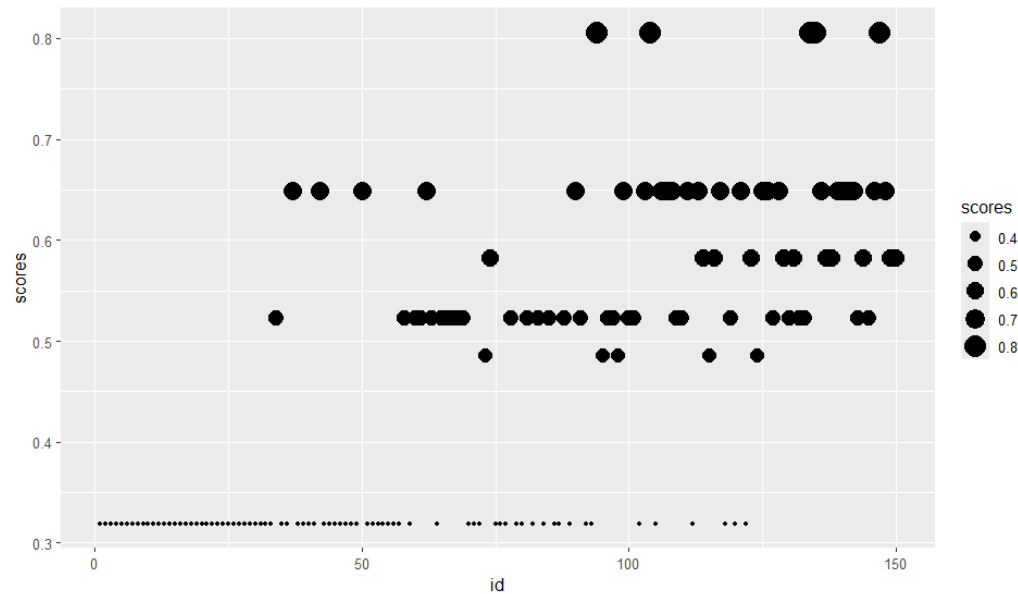


Newer synthetics



Unsupervised detection:

*Rare patterns and anomalies -
scores and posterior probabilities*



Thank you !

- violeta.calian@hagstofa.is
- <https://github.com/violetacln/testingSDCtools>
- <https://github.com/violetacln/SIPP>